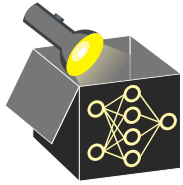
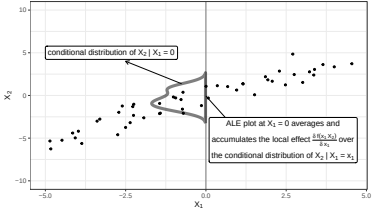
# Interpretable Machine Learning



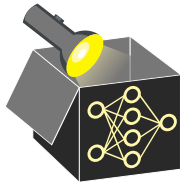
## Feature Effects
## Accumulated Local Effect (ALE) plot




**Learning goals**

- Understand ALE plots
- Difference between ALE and PD plots

# ACCUMULATED LOCAL EFFECTS (ALE) ▸ "Apley, Zhu" 2020

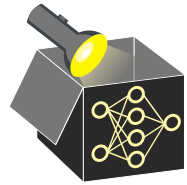


ALE plots estimate marginal effect of a feature by accumulating its local effects (integrating partial derivatives), evaluated in regions supported by the data.

**Computation Steps:**

1. **Estimate local effects** $\frac{\partial \hat{f}(x_S, \mathbf{x}_{-S})}{\partial x_S}$ (via finite differences)
   $\Rightarrow$ Removes unwanted main effects of other features $\mathbf{x}_{-S}$ (unlike M plots)

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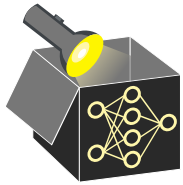


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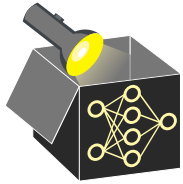


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2. **Average local effects** over conditional distr. $\mathbb{P}(\mathbf{x}_{-S}|x_S)$ similar to M plots
   $\Rightarrow$ Avoids extrapolation (unlike PD plots)
3. **Accumulate:** Integrate averaged local effects up to a specific $x \in \mathcal{X}_S$
   $\Rightarrow$ Reconstructs main effect of $x_S$

# FIRST ORDER ALE FUNCTION

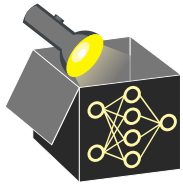


**Uncentered ALE Function** evaluated at $x \in \mathcal{X}_S$ (domain of feature $x_S$):

$$\tilde{f}_{S,\text{ALE}}(x) = \underbrace{\int_{z_0}^{x}}_{(3)} \underbrace{\mathbb{E}_{\mathbf{x}_{-S}|x_S=z_S}}_{\substack{(2)\text{ average} \\ \text{locally}}} \left( \underbrace{\frac{\partial \hat{f}(z_S, \mathbf{x}_{-S})}{\partial z_S}}_{(1)\text{ local effect}} \right) dz_S = \int_{z_0}^{x} \int \frac{\partial \hat{f}(z_S, \mathbf{x}_{-S})}{\partial z_S} d\mathbb{P}(\mathbf{x}_{-S} \mid z_S) dz_S$$

- $x_S$ is feature of interest, with minimum value $z_0 = \min(x_S)$
- $z_S$ is integration variable ranging over $\mathcal{X}_S$, used to evaluate local effects
- $\mathbf{x}_{-S}$ denotes all other features (complement of $S$)

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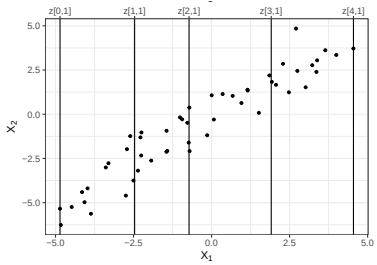
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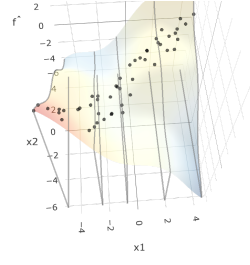
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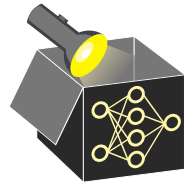
**Centering (to ensure identifiability):**

$$f_{S,\text{ALE}}(x) = \tilde{f}_{S,\text{ALE}}(x) - \underbrace{\int \tilde{f}_{S,\text{ALE}}(x_S)\, d\mathbb{P}(x_S)}_{\text{constant shift to mean zero}}$$

# ALE ESTIMATION: ILLUSTRATION



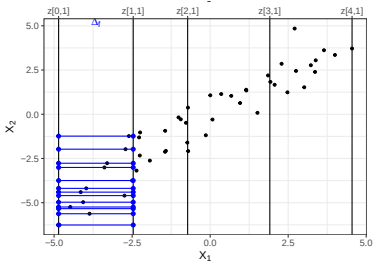



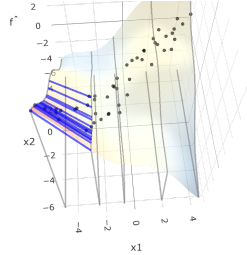
- **Motivation:** Partial derivatives are not well-defined for all models (e.g., tree-based methods). ⇒ Use finite differences within intervals instead.
- Partition the feature range of $x_S$ into $K$ intervals (vertical lines)
  - Define intervals:

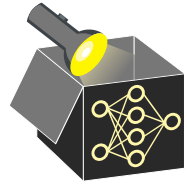
$$x_S \in [\min(x_S), \max(x_S)] \Rightarrow x_S \in [z_0, z_{1,S}] \cup [z_{1,S}, z_{2,S}] \cup \cdots \cup [z_{K-1,S}, z_{K,S}]$$

  - *Equidistant*: preserves resolution
  - *Quantile-based*: balances sample size per interval

# ALE ESTIMATION: ILLUSTRATION

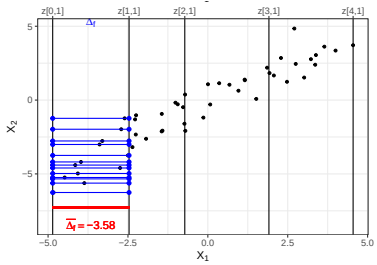


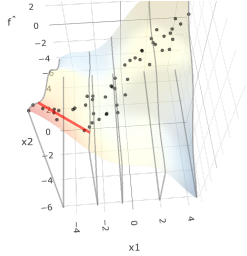



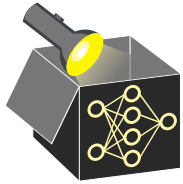



- For each observation in $k$-th interval, i.e., $\{i : x_S^{(i)} \in [z_{k-1,S}, z_{k,S}]\}$:
  - Replace $x_S^{(i)}$ with upper/lower interval bounds, keeping $\mathbf{x}_{-S}^{(i)}$ fixed
  - Compute obs.-wise finite difference of $i$-th obs. in $k$-th interval
    $\rightsquigarrow \hat{f}(z_{k,S}, \mathbf{x}_{-S}^{(i)}) - \hat{f}(z_{k-1,S}, \mathbf{x}_{-S}^{(i)})$ (approximates local effect)

# ALE ESTIMATION: ILLUSTRATION

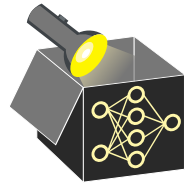


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- Average these finite differences over all observations in each interval
  $\rightsquigarrow$ Approximates **inner integral** $\mathbb{E}_{\mathbf{x}_{-S}|x_S=z_S}\left[\partial \hat{f}/\partial z_S\right]$
- Accumulate these averages from $z_0$ to the point of interest $x \in \mathcal{X}_S$
  $\rightsquigarrow$ Approximates **outer integral** over $z_S \in [z_0, x]$
  $\Rightarrow$ uncentered ALE function

# ALE ESTIMATION: FORMULA

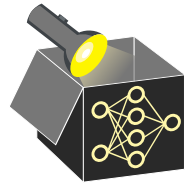


**Estimated uncentered ALE:** For a point $x \in \mathcal{X}_S$, define:

$$\hat{\tilde{f}}_{S,ALE}(x) = \sum_{k=1}^{k_S(x)} \frac{1}{n_S(k)} \sum_{i:\, x_S^{(i)} \in [z_{k-1,S}, z_{k,S}]} \left[ \hat{f}(z_{k,S}, \mathbf{x}_{-S}^{(i)}) - \hat{f}(z_{k-1,S}, \mathbf{x}_{-S}^{(i)}) \right]$$

- $[z_{k-1,S}, z_{k,S}]$: $k$-th interval of feat. $x_S$ with interval bounds $z_{k-1,S}$ and $z_{k,S}$
- $k_S(x)$: index of the interval in which $x$ lies
- $n_S(k)$: number of observations in interval $k$
- $\mathbf{x}_{-S}^{(i)}$: all other features held fixed for $i$-th observation

# ALE ESTIMATION: FORMULA



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**Centering:** Ensure identifiability by subtracting mean uncentered ALE ($c$):

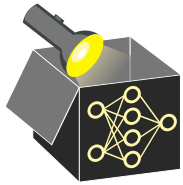
$$\hat{f}_{S,\text{ALE}}(x) = \hat{\tilde{f}}_{S,\text{ALE}}(x) \; - \; c, \qquad c = \tfrac{1}{n} \textstyle\sum_{i=1}^{n} \hat{\tilde{f}}_{S,\text{ALE}}\left(x_S^{(i)}\right).$$

**Efficient centering (used in implementations):** Use weighted trapezoidal averaging of interval-wise boundary values (avoids redundant re-evaluation at all $n$ points):

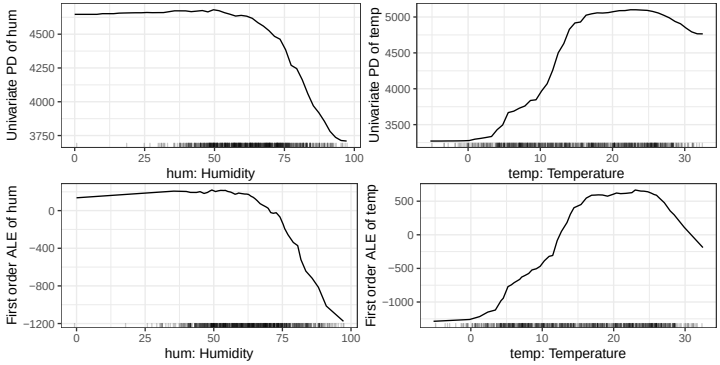
$$c = \textstyle\sum_{k=1}^{K} \frac{1}{2} \cdot \left(\hat{\tilde{f}}_{S,\text{ALE}}(z_{k-1,S}) + \hat{\tilde{f}}_{S,\text{ALE}}(z_{k,S})\right) \cdot \frac{n_S(k)}{n}$$

**Plotting:** Visualize pairs $\left(z_{k,S}, \hat{f}_{S,\text{ALE}}(z_{k,S})\right)$ for all interval boundaries $z_{k,S}$.

# BIKE SHARING DATASET: FIRST ORDER ALE

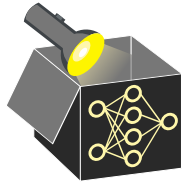


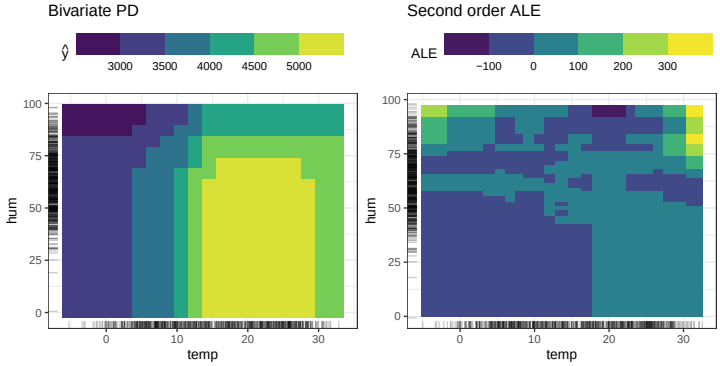
- **Visual comparison:** PD plot (top) vs. First-order ALE plot (bottom)
- **Shape:** Similar trends in both plots; $y$-axis scale differs due to centering
- **Interpretation:** ALE accounts for feature dependencies and avoids extrapolation into unsupported regions
  - $\rightsquigarrow$ PD reflects model behavior in entire feature space ("true to the model")
  - $\rightsquigarrow$ ALE focuses on effects in data-supported regions ("true to the data")



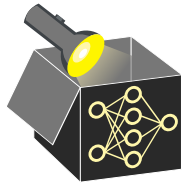
# BIKE SHARING DATASET: SECOND ORDER ALE

Unlike bivariate PD plots, 2nd-order ALE plots only estimate pure interaction between two features (1st-order effects are not included).





# PD VS. ALE



**PD:**

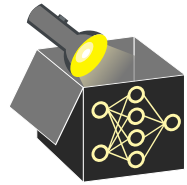
$$f_{S,PD}(x_S) = \mathbb{E}_{\mathbf{x}_{-S}}\left(\hat{f}(x_S, \mathbf{x}_{-S})\right)$$

**ALE:**

$$f_{S,\text{ALE}}(x) = \int_{z_0}^{x} \mathbb{E}_{\mathbf{x}_{-S}|x_S=z_S}\left(\frac{\partial \hat{f}(z_S, \mathbf{x}_{-S})}{\partial z_S}\right) dz - \text{const}$$

- Recall: PD directly averages predictions over marginal distribution of $\mathbf{x}_{-S}$
- ALE is faster: $O(2 \cdot n)$ model calls vs. $O(n \cdot g)$ for PD with $g$ grid points
- Difference 1: ALE averages
  - prediction changes (via partial derivatives, estimated by finite differences)
  - over conditional distribution $\mathbb{P}(\mathbf{x}_{-S}|x_S = z_S)$

# PD VS. ALE



**PD:**

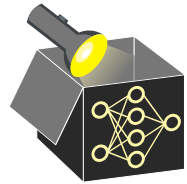
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- Difference 2: ALE integrates these partial deriv. over $z_S \in [z_0, x] \subseteq \mathcal{X}_S$
  $\rightsquigarrow$ isolates effect of $x_S$ and removes main effect of other dependent feat.

# PD VS. ALE



**PD:**

$$f_{S,PD}(x_S) = \mathbb{E}_{\mathbf{x}_{-S}}\left(\hat{f}(x_S, \mathbf{x}_{-S})\right)$$

**ALE:**

$$f_{S,\text{ALE}}(x) = \int_{z_0}^{x} \mathbb{E}_{\mathbf{x}_{-S}|x_S=z_S}\left(\frac{\partial \hat{f}(z_S, \mathbf{x}_{-S})}{\partial z_S}\right) dz - \int \tilde{f}_{S,\text{ALE}}(x_S)\, d\mathbb{P}(x_S)$$

- Recall: PD directly averages predictions over marginal distribution of $\mathbf{x}_{-S}$
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- Difference 1: ALE averages the
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- Difference 2: ALE integrates these partial deriv. over $z_S \in [z_0, x] \subseteq \mathcal{X}_S$
  $\rightsquigarrow$ isolates effect of $x_S$ and removes main effect of other dependent feat.
- Difference 3: ALE is centered so that $\mathbb{E}_{x_S}\left(f_{S,ALE}(x)\right) = 0$